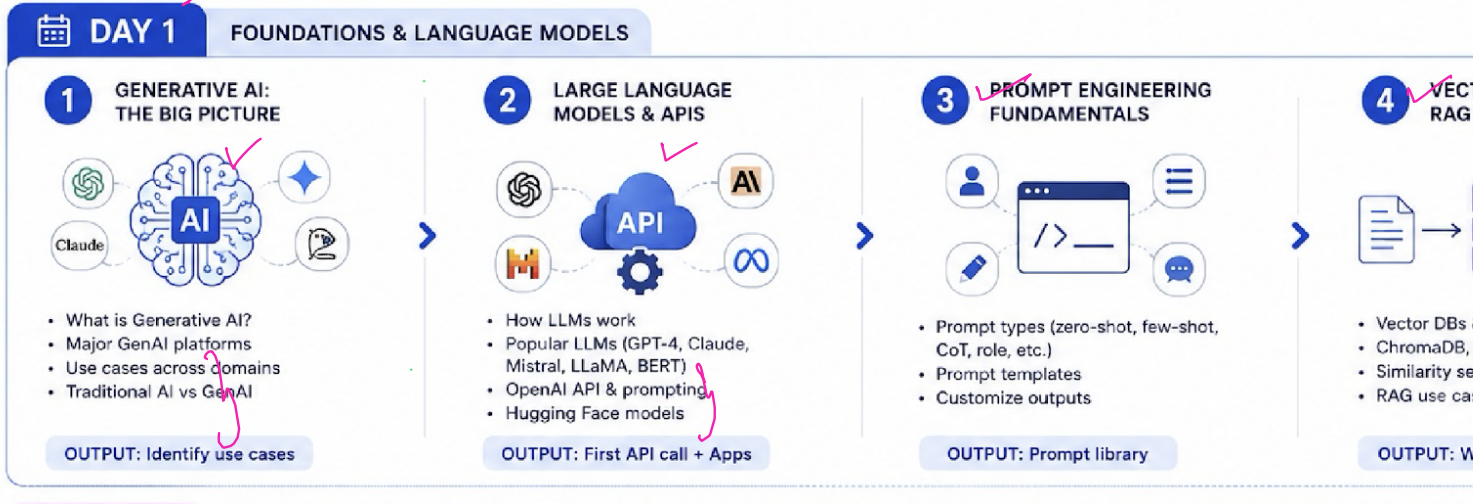


# Day 1

Fri 11 Sep 2026 1:53 PM

AI LML > DL > Word Embed LLMs



AI → Artificial Intelligence → CS → Machine

↳ 70-80%

↳ Human

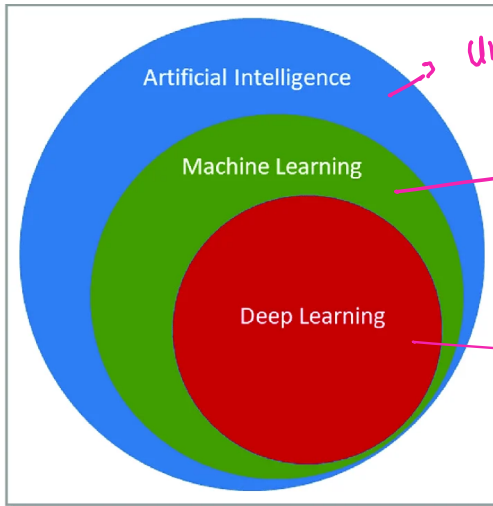
↳ RAG

↳ Algo

Weak AI - Mimic

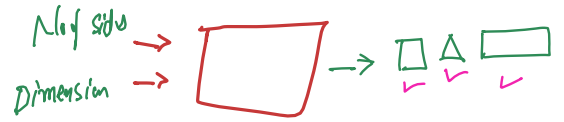
General AI → Intelligence

Strong AI → Surpassing human → Movies



Umbrella

without explicit programs



Programming

Explicitly mentioning

No-4  
Dim - 60 > 50

Data →

↳ Algorithm - ✓

↳ Patterns → Output 0.80%

Independent

Input  
Input 'X'

Target 'Y'

Dependent

| 1 | sepal_length ✓ | sepal_width ✓ | petal_length ✓ | petal_width ✓ | species ✓ |
|---|----------------|---------------|----------------|---------------|-----------|
| 2 | 5.1            | 3.5           | 1.4            | 0.2           | setosa    |
| 3 | 4.9            | 3.0           | 1.4            | 0.2           | setosa ✓  |
| 4 | 4.7            | 3.2           | 1.3            | 0.2           | setosa    |
| 5 | 4.6            | 3.1           | 1.5            | 0.2           | setosa    |
| 6 | 5.0            | 3.6           | 1.4            | 0.2           | setosa    |
| 7 | 5.4            | 3.9           | 1.7            | 0.4           | setosa    |

Class A  
Square

Class B  
Triangle

→ Supervised → Label + Data → Model → Predict

→ Unsupervised → Data → Model - Classify / Clustering

→ Reinforcement → feedback

↳ Predict - Reward / Penalty

Unsupervised

→ Data ✓

→ No

→ Explicit - No

→ Pattern / Structure  
↳ Clusters

Reinforcement

interaction with

No

Explicitly

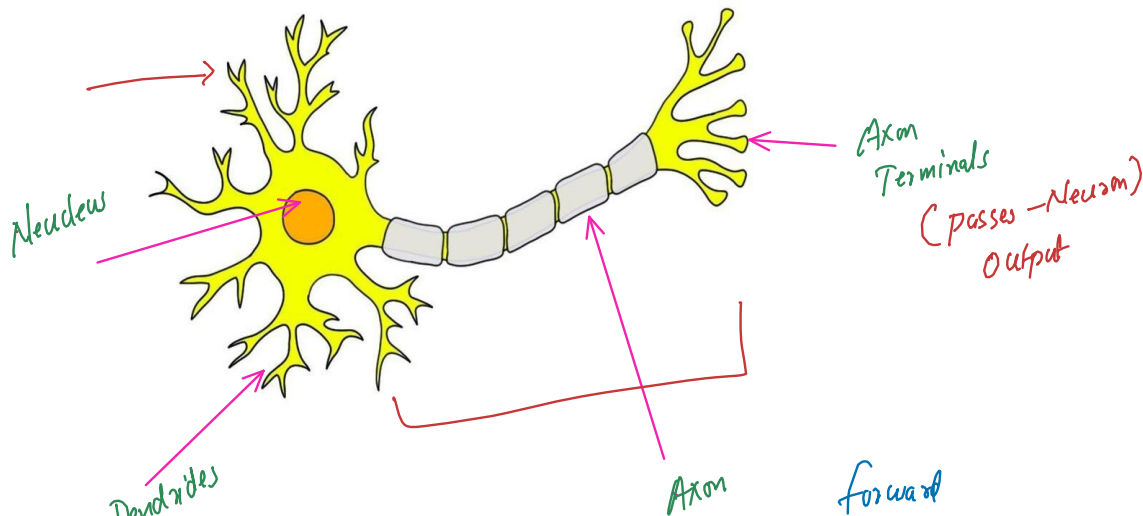
Sequence data

↳ Robot

# Deep learning

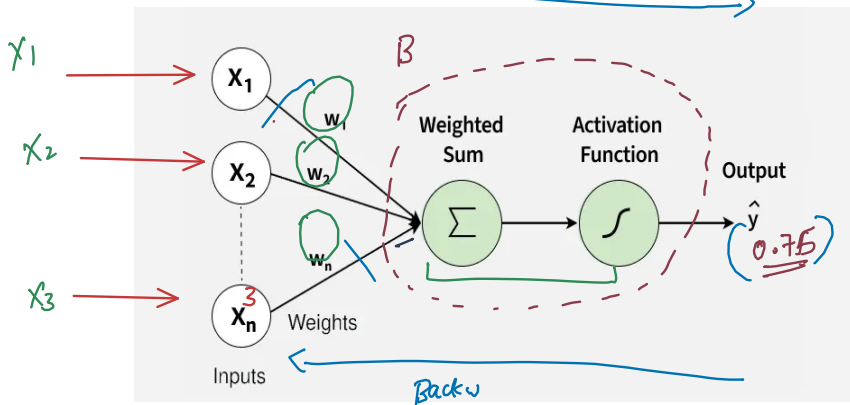
↳ subset of ML

## Neuron - Biological



1 Epoch

forward



$$Y = 1$$

$$\hat{Y} = 0.65$$

$$T = 0.85$$

$$L = (y - \hat{y})^2$$

$$= (1 - 0.65)^2$$

$$= 0.35$$

$E \leq \begin{matrix} A \\ P \\ TS \end{matrix}$

P

- ✓ Weights → importance > 0.01 → Randomly
- ✓ Bias → Starting, prevents the neuron - dead

Weighted Sum:

$$f(x) = X \cdot w + B - \text{Bias}$$

$$= 0.7 \times 0.05 + 0.01$$

$$= 0.75 \rightarrow$$

Activation

Relu

$$0.75$$

Activation

↳ Neuron / Inactive

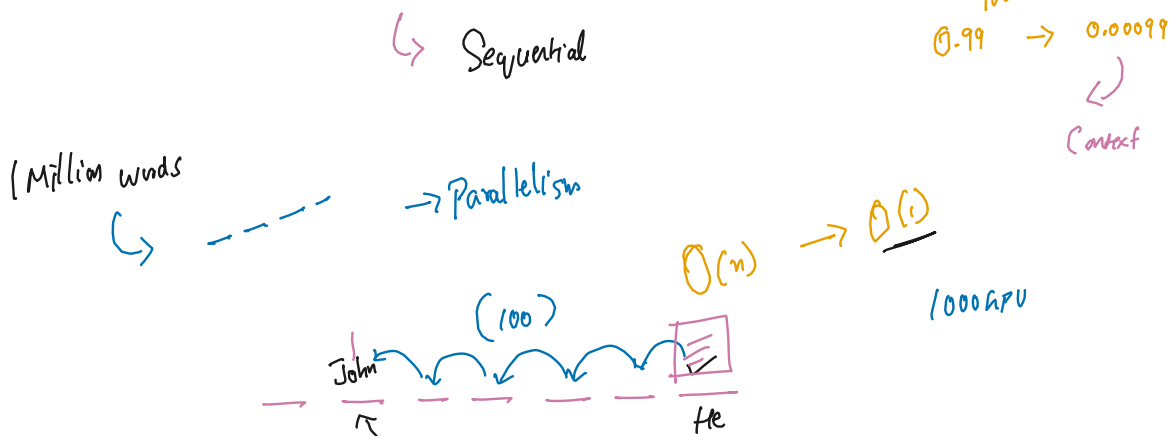
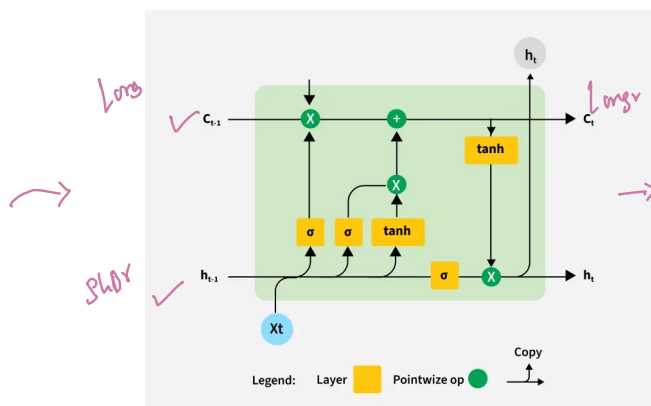
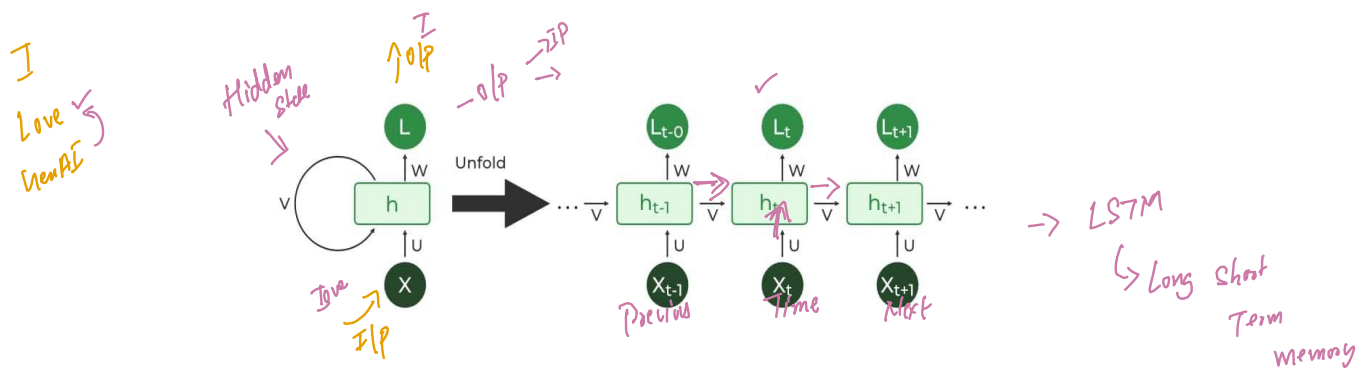
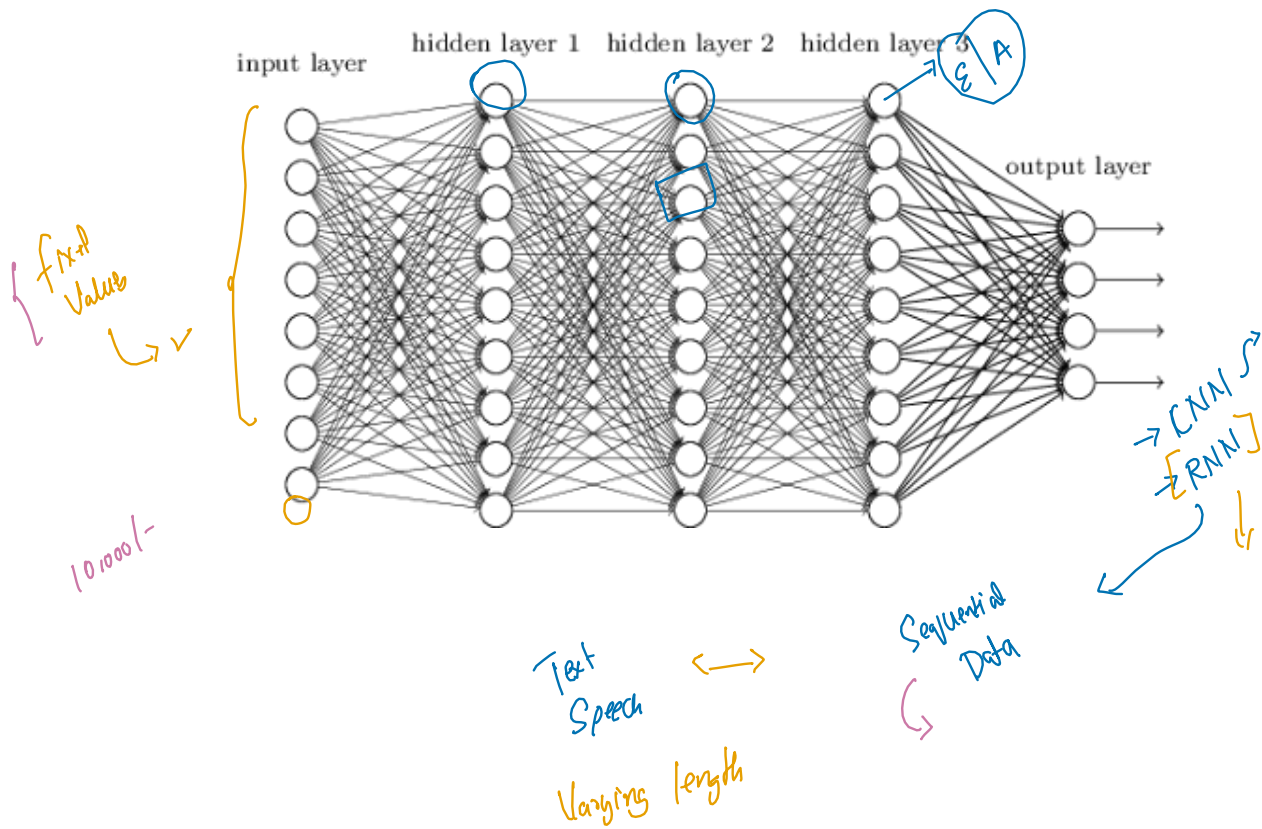
Relu

ILP

↑ve - +ve  
-ve - 0

→ 0-1

learned Parameters



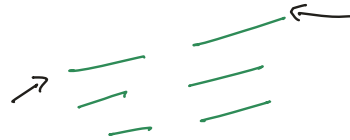
# ↓ Affection Mechanism

Words → Presented to PC

Translator

f - E

Lookup



Binary

Programme → Compiler → ML

① OHE

" I Love Internet , it's awesome "

ML

|   |       |   |   |   |   |   |   |          |
|---|-------|---|---|---|---|---|---|----------|
| ✓ | ✓     | 1 | 0 | 0 | 0 | 0 | 0 | → Binary |
| ✓ | Love  | 0 | 1 | 0 | 0 | 0 | 0 |          |
| ✓ | Int   | 0 | 0 | 1 | 0 | 0 | 0 | ✓        |
| ✓ | it's  | 0 | 0 | 0 | 1 | 0 | 0 |          |
| ✓ | awsme | 0 | 0 | 0 | 0 | 1 | 0 |          |

10,000 words

Word → [ 1 9999 0 ]

↳ Sparse Vector

↳ Storage

Semantic Understanding

Friend → Hi

far - close

Bye

Greeting

> Semantic

leave

ss

Holiday

# Embeddings

→ Numerical Represent

↳ Sparse - (X)

→ Dense ✓

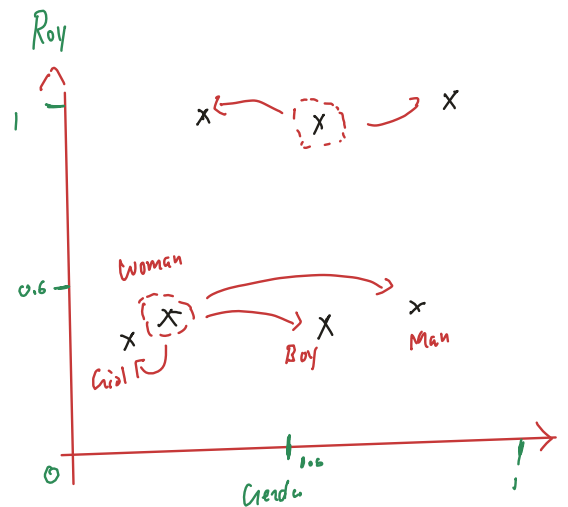
King →  $\begin{matrix} \text{Wealth} & \text{Status} & \text{Kind} & \text{Developer} \\ \hline [0.80, 0.90, 0.19, 0.76] \end{matrix}$  → Dimensions  
Vector Embedding

Royalty

Gender

0 0.5 1  
f M  
0 0.5 1

|        |      |      |
|--------|------|------|
| King   | 0.95 | 0.85 |
| Queen  | 0.95 | 0.4  |
| Man    | 0.55 | 0.75 |
| Woman  | 0.45 | 0.25 |
| Prince | 0.90 | 0.65 |
| Boy    | 0.40 | 0.65 |
| Girl   | 0.40 | 0.15 |



2 Dimensions → Understanding

$[384, 512, 768, 1024, 1636, 3096, 12000 +]$  → Embedding Models  
↳

→ All mini 16 v2 → 60MB → 384

→ Text embedding 3 small → OpenAI → 1536

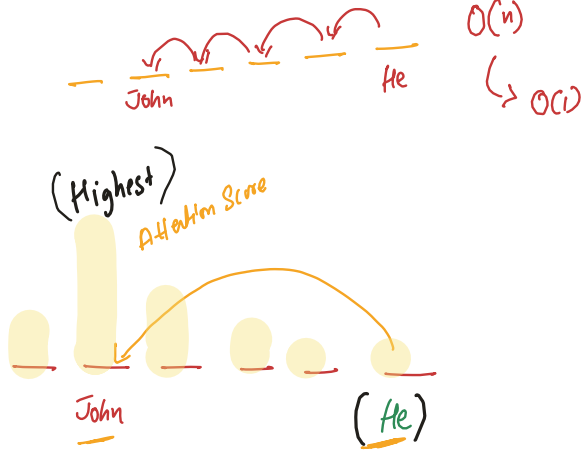
Text → Embedding → Vector

Bank — financial  
— River

Orange — color  
— fruit

Sentence → Sentence embedding

## Attention Mechanism



Q, K, V  
Query Key Value

"Attention is all you need"

2017  
→ Vaswani  
(F → E)  
512 Dimension

A - A      A - B, C  
B - B      ⇒ B → C, A  
C - C      C → A, B

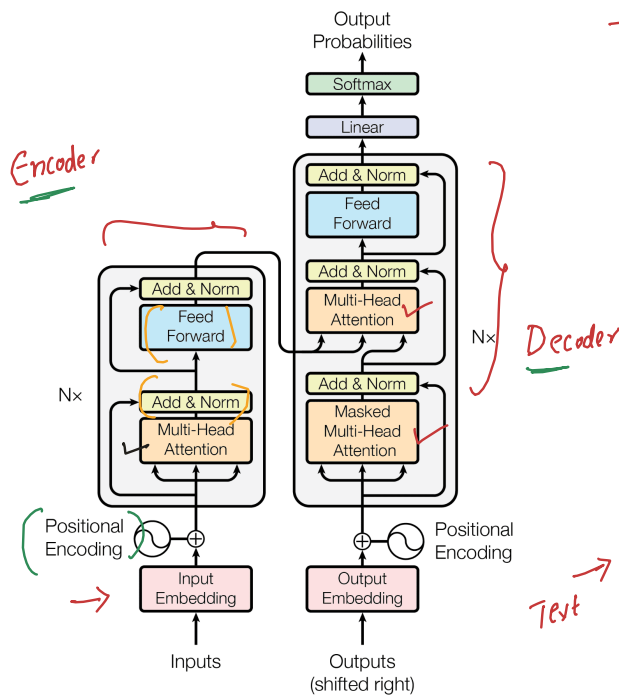
A → B - 70%  
C - 30%

## Transformer

12 - Attention

8 → 512

Dog Bitn Man  
Man Bite Dog ⊕



Transformer

12 \* 200 + 96 Trans

Decoder

Encoder → BERT  
Decoder → GPT, Gemini, Claude  
Test En-dec  
↳ T5 - Man  
Translation

GenAI

↳ Generating New Content

Poex

Image

Text

(LLM)

large

language

Model

↳ Mathematical

↳ learned parameters

↳ Entire

↳ NL - NL

LLMs

Gemini  
GPT  
Claude  
Opus  
Deepseek  
Ollama

de

Closed Weight

Open Weight Models

Weights - Not Available

GPT  
Claude  
gemini  
Grok

Qwen  
mistral  
GLM

Open Weightrs

Qwen  
Mistral  
LFM  
Falcon  
Deepseek

## Real-World Business Scenario

## LLM Model

**E-commerce company:** Analyze thousands of customer complaints, identify the root cause of returns, summarize recurring product issues, and generate recommendations for the product team.

**Bank:** Build an AI assistant that reviews customer queries, checks account-related information through APIs, explains transactions, and escalates suspicious or complex cases to human agents.

**Healthcare company:** Analyze patient documents and reports, extract key information, summarize the case for doctors, and answer questions based only on the organization's approved medical knowledge base.

**Software company:** Give developers an AI coding assistant that understands a large existing codebase, identifies bugs, generates code, explains legacy code, and helps plan complex application changes.

**Manufacturing company:** Analyze machine logs, maintenance reports, sensor data, and technician notes to identify possible equipment failures and recommend the next troubleshooting steps.

## Prompt Engineering

language → LLM

Prompt → Precede  
↳ Response

intent → Bridge → LLM

## Component

- ✓ ① Task — Work Summarise Transformer
  - ✓ ② Instructions — How-way — 2 points → 5th grade → 5 point
  - ✓ ③ Context → Around → Tech Conference
  - ✓ ④ Input → file, txt
- Q: What is 0  
A: or four

|   |                                 |                                                          |                                                                      |
|---|---------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
| 1 | Zero-Shot Prompting             | Ask directly without examples                            | "Summarize this report in 5 bullets."                                |
| 2 | Few-Shot Prompting              | Give examples to show the expected pattern               | "Positive → Good. Negative → Bad. 'Excellent service' → ?"           |
| 3 | Role Prompting                  | Assign a role/persona                                    | "Act as a senior financial analyst..."                               |
| 4 | Instruction Prompting           | Clearly specify what the model should do                 | "Extract the customer name, issue, and priority."                    |
| 5 | Contextual Prompting            | Provide relevant background/context                      | "Our company sells SaaS products to banks. Based on this context..." |
| 6 | Chain-of-Thought / Step-by-Step | Ask the model to reason through a problem systematically | "Solve the problem step by step and provide the final answer."       |
| 7 | Structured Output               | Define the format of the expected answer                 | "Return JSON with name, email, and company."                         |
| 8 | Constraint Prompting            | Specify boundaries/rules                                 | "Answer in less than 100 words and use simple English."              |

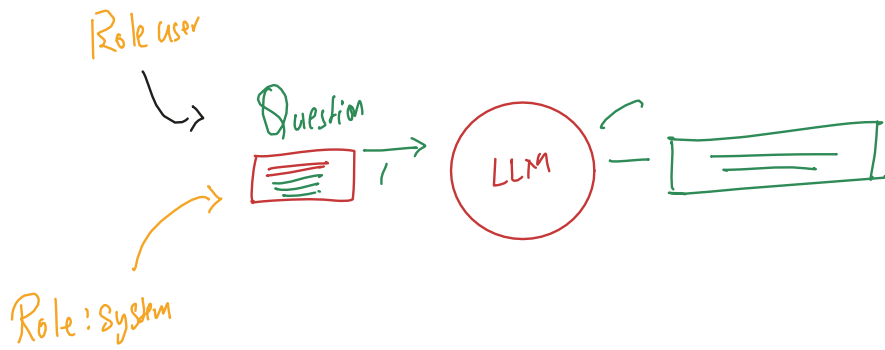
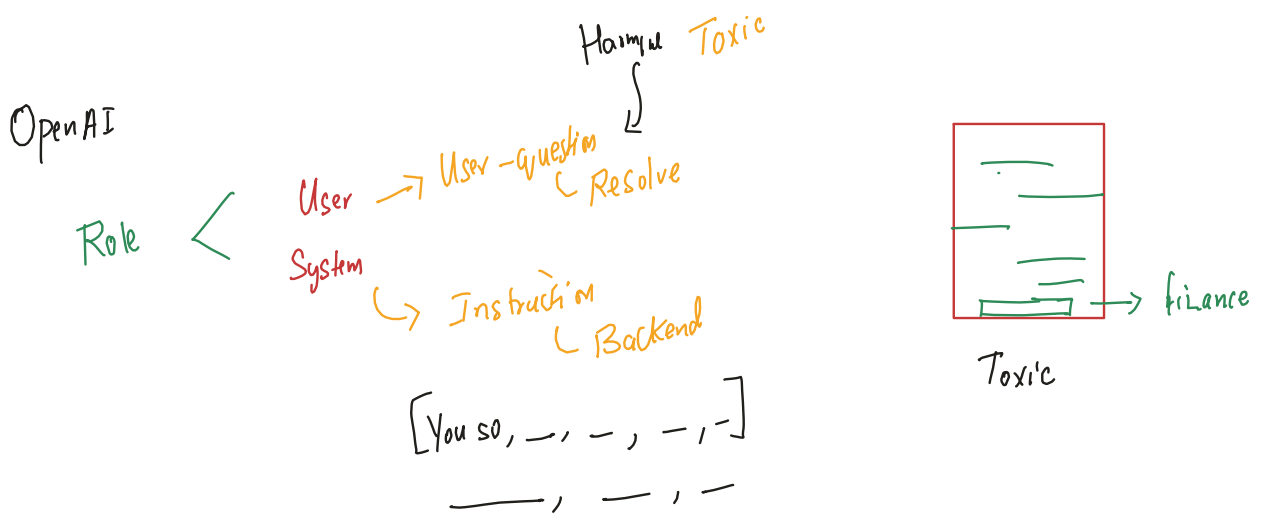
Q: What  
A: —  
[ — ]

Self Critique → 100 years ← Eliminate  
↳ 10

Meta → LLM - Prompting



↳ Meta

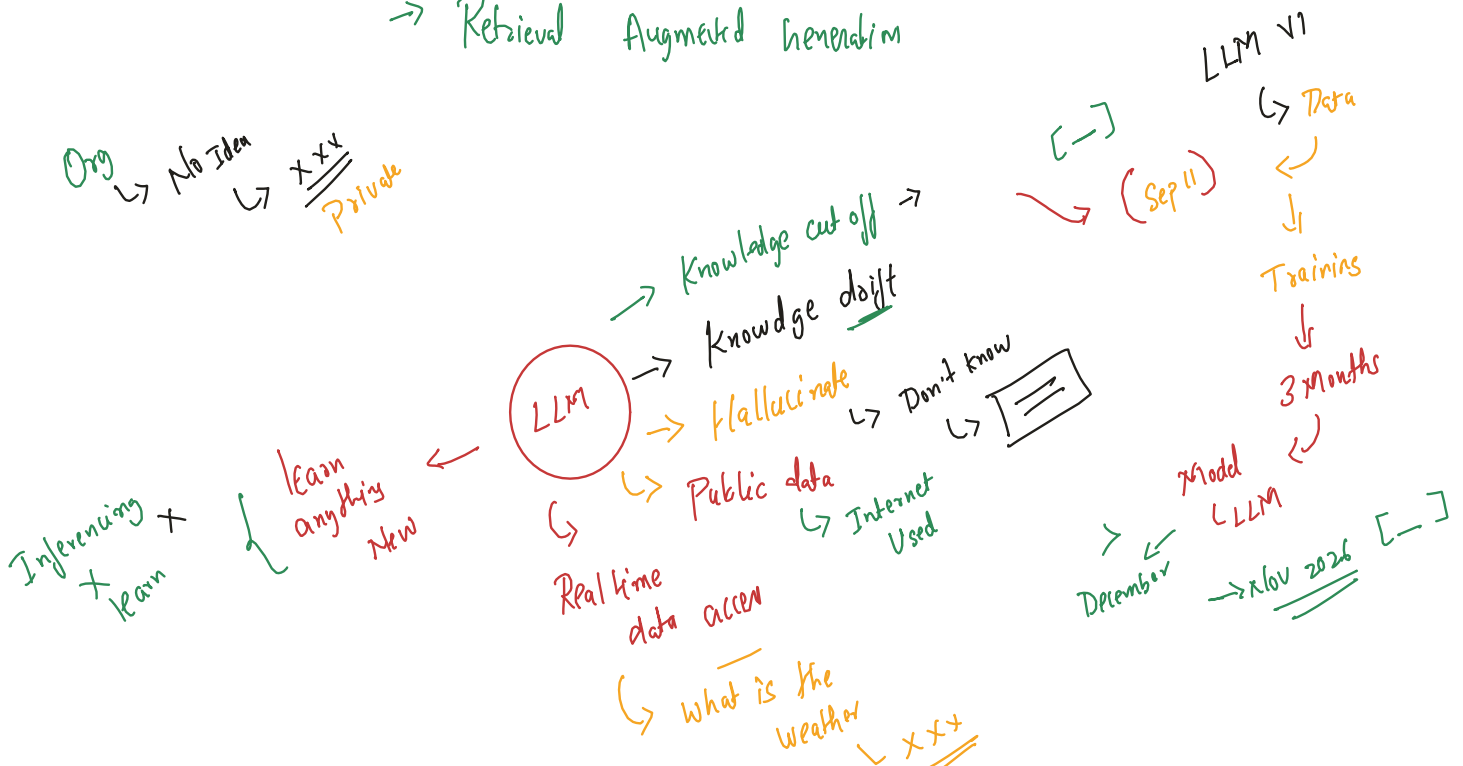


LLM - Gpt 40 Mini

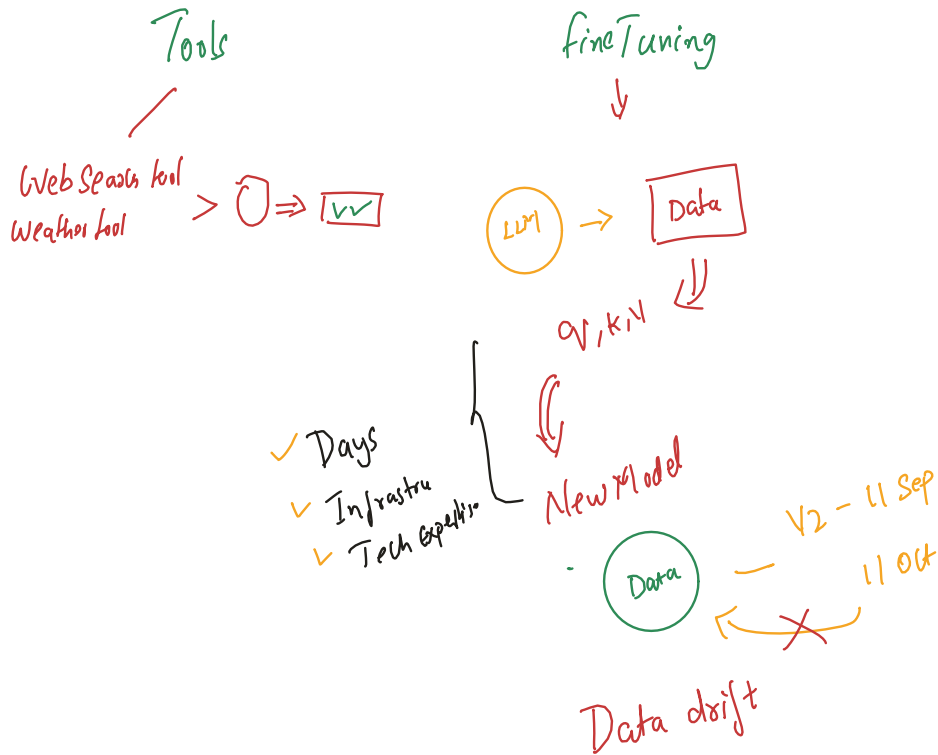
API - Call ----> My Name Is Ahmed

API Call - ----> What's My Name?

RAG → Retrieval Augmented Generation



Limitations  $\rightarrow$  Org  $\Rightarrow$  Public Data, Hallucinate, Reason - [ ]  
(DB(x))  
Source  $\leftarrow$  Citation



RAG

LLM - Aside  
(Around it  $\rightarrow$  LLM)

Retrieval Augmented Generation

Query  $\rightarrow$  LLM  $\rightarrow$  Answer

Query  $\rightarrow$  Retrieval KB  $\rightarrow$  LLM  $\rightarrow$  Answer

### 3 Stages

#### ① Data Ingestion

##### ① Load Data

Pdf, docx  
JSON  
web, XL, PPTx  
> Read  $\Rightarrow$  Data Loaders

Pdf - Pdf Loader  
CSV - CSV Loader  
Web  $\rightarrow$  Web Loaders  
} Loaders  $\rightarrow$  Valid  
Data

##### ②

1000 Pages  
Answer

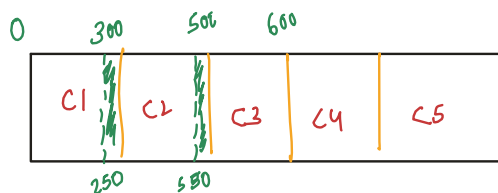
$\rightarrow$  No Page Sections Chapters

Question -  $\langle \Rightarrow \rangle$   
Answers

? Ahmed  $\rightarrow$  Page No  
Sections  
Paragraph

Sentences  
Chapters

#### Chunking

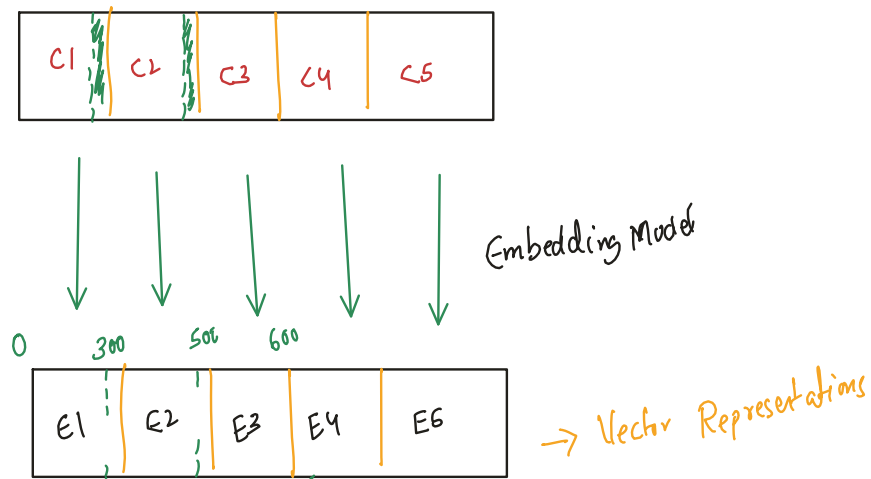


fixed = 300 Tokens  
overlap = 50 Tokens

✓ Sentence  
Paragraph  
Section  
fixed size  
fixed chunking - overlap  
Semantic chunking  
Recursive chunking

##### ③

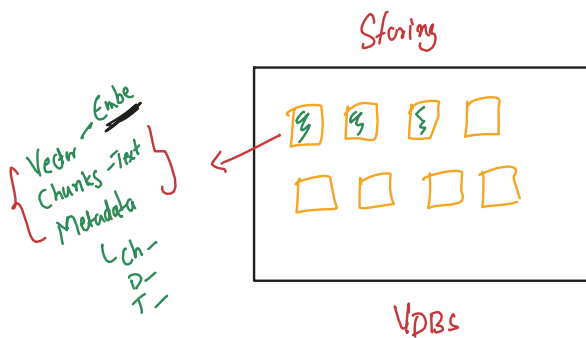
#### Embedding



## ④ Vector DBs

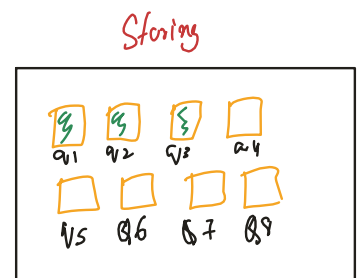
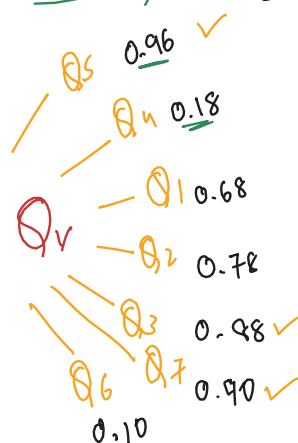
10,000 - Vectors Embedding

↳ Database  
↳ VDBs



→ ChromaDB  
→ Pinecone  
→ FAISS  
→ Milvus  
→ PG Vector

Raw Document → Structured / DBs



VDBs

Top k=3

## ② Retrieval

Question  $\rightarrow$  Embedding Model  $\rightarrow$  Embedding Vector  $\rightarrow$  Q<sub>v</sub>

Text  $\rightarrow$   $\begin{matrix} \checkmark \\ Q_1 \\ \downarrow \\ C_1 \end{matrix}$   $\begin{matrix} \checkmark \\ Q_2 \\ \downarrow \\ C_3 \end{matrix}$   $\begin{matrix} \checkmark \\ Q_6 \\ \downarrow \\ C_5 \end{matrix}$

Text  $\rightarrow$   $\begin{matrix} Q_r \\ Q \end{matrix}$

3

Generation

$\begin{matrix} C_1 & C_3 & C_5 \\ \text{[ Your, RAG, chunks ]} \end{matrix}$   $\Rightarrow$   $\boxed{\text{=}}$   
Q

$\boxed{\text{=}}$

LLM

$\hookrightarrow$  Your Own documents  
 $\hookrightarrow$  Traced

$\rangle$  Reliable

Fixed Sized

small

[ ]

Question → Agentic RAG  
 LLM  
 own → RAG

Harsh Kumar 8:14 PM  
 Do you as user create the chunks? - Yes  
 When would you use what chunking strategy? → [ ] - Sentence  
 → Medical → Semantic  
 Harsh Kumar 8:30 PM  
 How does it determine that if needs to kick off a retrieval or simply just submit the request to the llm? → LLM → Last flow  
 Rohit Sharma 8:32 PM  
 Aren't we sharing our internal data with the LLM in this approach? Wouldn't this create potential data security and privacy concerns? → Yes - LLM  
 → Providers

Private Instance

→ openAI  
 Private

Local LLMs

→ [X] [X]

$$\cosine = \frac{S1 \cdot S2}{|A| \cdot |B|}$$
  
 = Angle A.B  
 → Embeddings

Max Token

→ Cat - - - - -  
 ↓  
 Embedding - 1000 Dim  
 ↓  

|    |          |
|----|----------|
| S1 | [ 1000 ] |
| S2 | [ 1000 ] |

Suneel Musunuru 8:33 PM Edited  
 Chunks are bigger compared to user queries. , How does similarity comparison between chunk embeddings and embeddings of short, focused user queries works?  
 Nikhil Kale 8:35 PM  
 textual chunking can be based on sentences, paragraphs etc. does chunking also work for images , tables, complex pdf formats.  
 documents can contain references to other documents, sections, are there ways to handle such complexity.

→ Pdf - Tables  
 - Handr , images  
 Unstructured

lamaIndex

→ OCR - Text  
 → Image → LLM  
 ← description